

Logistic Regression

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Preface

This ebook documents a complete logistic regression workflow, implemented across four GitHub repositories by Roger J. DeAngelis. The series demonstrates a systematic approach to:

1. Optimum binning of both character and numeric covariates in preparation for logistic regression (Chapter I)
2. Identifying the best five logistic models from all possible combinations of predictors (Chapter II)
3. Verifying the training logistic model using a holdout sample (Chapter III)
4. Extracting and presenting key outputs from the final logistic model (Chapter IV)

The methodology emphasizes interpretability, validation, and practical deployment. All code and examples are provided in SAS, with extensive use of macros and reproducible workflows.

The repository text throughout this ebook is presented in a fixed font for clarity.

I suggest readers use this ebook to acquaint themselves with the process, but spend more time implementing workflow using the four repositories below.

GitHub: <https://github.com/rogerjdeangelis/utl-altair-slc-chapter-I-optimum-binning-in-preparation-for-logistic-regression>

GitHub: <https://github.com/rogerjdeangelis/utl-altair-slc-chapter-II-identifying-the-best-five-logistic-models>

GitHub: <https://github.com/rogerjdeangelis/utl-altair-slc-chapter-III-verifying-training-logistic-using-holdout>

GitHub: <https://github.com/rogerjdeangelis/utl-altair-slc-chapter-IV-key-outputs-from-the-final-logistic-model>

CHAPTER I: OPTIMUM BINNING IN PREPARATION FOR LOGISTIC REGRESSION

Repository: [utl-altair-slc-chapter-I-optimum-binning-in-preparation-for-logistic-regression](https://github.com/rogerjdeangelis/utl-altair-slc-chapter-I-optimum-binning-in-preparation-for-logistic-regression)

GitHub: <https://github.com/rogerjdeangelis/utl-altair-slc-chapter-I-optimum-binning-in-preparation-for-logistic-regression>

I. Analysis Overview

The first chapter focuses on the critical preprocessing step of **optimum binning** - grouping continuous and categorical variables into bins that exhibit monotonic relationships with the binary response variable. This step is essential for:

- Ensuring that the logistic model satisfies the **linearity assumption** for continuous predictors
- Creating **interpretable** categorical predictors with meaningful odds ratios
- Reducing noise and overfitting by collapsing sparse categories
- Improving model **stability** and performance

The binning process uses Mantel-Haenszel chi-square statistics to determine optimal cut points that maximize the association between each predictor and the response.

II. Campaign Strategy

The campaign strategy involves:

1. Data validation and verification of raw input
2. Handling missing values - converting all missing formats to a consistent '?' representation
3. Dropping low-cardinality variables (one-to-one relationships)
4. Optimizing variable length to the longest observed value in the data
5. Creating holdout and training tables (70,000 training / 30,000 holdout)

III. Key Output - Sample of Binning One Covariate

The binning process produces two key Excel reports:

Report	Description	Link
Character Binning Report Numeric	Binned character covariates with chi-square & MH stats	lgs_MgmChrCut.xl
Binning Report	covariates with chi-square & MH stats	lgs_MgmNumCut.xl

Each report contains:

- Bin boundaries for each covariate
- Odds ratios per bin (compared to a reference bin)
- Chi-square statistics and p-values
- Mantel-Haenszel trend tests for monotonic association

IV. Interpreting Binned Data

The binned data is represented in two forms:

1. Normalized (long and skinny) format - each observation is expanded to multiple rows (one per bin variable), with just 5 key variables:
 - **_ID** (identifier)
 - **_NAME** (variable name)
 - **_VALUE** (bin value)
 - **_RESPONSE** (binary outcome)
 - **_WEIGHT** (optional)

2. Denormalized (wide) format - each original observation has one row with 30-38 binned indicator variables

Character variables: 14 original -> 40 bin indicators (denormalized)

Numeric variables: 13 original -> 46 bin indicators (denormalized)

V. Input

The raw input data is provided as:

- `lgs_raw.zip` - containing the raw training and holdout datasets

https://github.com/rogerjdeangelis/utl-altair-slc-chapter-I-optimum-binning-in-preparation-for-logistic-regression/blob/main/lgs_raw.zip

VI. Output - Five Tables

The chapter produces five key output tables:

Table	Description	Format
<code>lgs_mgmAllChrNum</code>	Final logistic-ready binned training data (wide) SAS dataset	Normalized character binned data (980k rows) SAS dataset
<code>lgs_mgmNrmChr</code>	Normalized numeric binned data (910k rows) SAS dataset	Character binning summary (40 obs, 14 vars) Excel
<code>lgs_mgmNrmNumSrt</code>	Numeric binning summary (46 obs, 13 vars) Excel	
<code>lgs_mgmChrCutRpt</code>		
<code>lgs_mgmNumCut</code>		

VII. Theory

The binning algorithm is based on the principle of maximum association:

- For each predictor, the algorithm searches for cut points that maximize the chi-square statistic for the 2xk contingency table (response x bins)
- Mantel-Haenszel chi-square is used to test for linear trend across ordered bins
- The optimal number of bins is determined by balancing:
 - Statistical significance (maximizing chi-square)
 - Interpretability (minimizing the number of bins)
 - Monotonicity (ensuring odds ratios increase or decrease consistently)

VIII. Method - Program Steps

The complete program consists of ten major steps:

1. Contents of raw input data
2. Validate and verify raw data; fix issues with raw data
3. One missing value - Drop one-to-one variables; very low cardinality variables. Convert the many missing variable formats to just '?'
4. Optimize variable length to the longest in the data
5. Create Holdout and Training Tables
6. Bin the character variables in groups with common odds ratios

- Create normalized (long and skinny binned data with just 5 variables)
 - Create denormalized (wide binned table with 30 variables)
7. Create Excel character data summary bin report with chi-square and Mantel-Haenszel stats
 8. Bin the numeric variables in groups with common odds ratios
 - Create normalized (long and skinny binned table with just 5 variables)
 - Create denormalized (wide binned table with 38 variables)
 9. Create Excel numeric data summary bin report with chi-square and Mantel-Haenszel stats
 10. Join raw training table, numeric binned data with character binned data for analysis

Macros Required (place in autocall library):

<https://github.com/rogerjdeangelis/utl-macros-used-in-many-of-rogerjdeangelis-repositories>

- utlflkil
- slc_optimalbin.sas (also in this repository)

https://github.com/rogerjdeangelis/utl_optlen

- utloptlen.sas

<https://github.com/rogerjdeangelis/voodoo>

- slc_voodoo20251126.sas (documentation included)

IX. Contents of Future Chapters

The subsequent chapters in this series cover:

Chapter	Topic
II III IV	Identifying the best five logistic models Verifying training logistic using holdout Key outputs from the final logistic model

Key outputs from the full series include:

- lgs_mgmFinalLogisticDiag - Logistic model diagnostics
- lgs_MgmGainsChart - Gains chart
- lgs_mgmTopChiValues - Most influential variables
- lgs_mgmTopIndexValues - Highest response variables
- lgs_MgmTopTen - List of top 12 scores
- lgs_MgmVenn - Comparison of covariate contribution to top percentile
- Final PDF presentation (slide deck)

Final logistic-ready output:

https://github.com/rogerjdeangelis/utl-altair-slc-chapter-I-optimum-binning-in-preparation-for-logistic-regression/blob/main/lgs_mgmallc_hnum

CHAPTER II: IDENTIFYING THE BEST FIVE LOGISTIC MODELS

Repository: `utl-altair-slc-chapter-II-identifying-the-best-five-logistic-models`

GitHub: <https://github.com/rogerjdeangelis/utl-altair-slc-chapter-II-identifying-the-best-five-logistic-models>

I. Preparation

Before running the analysis, two preparation steps are required:

1. Library statement - Add this to your autoexec: `libname workx "d:/wpswrkx";`
2. Macro - Copy `utl_mdlgetpos.sas` to your autocall library

II. Contents

All Excel files referenced in this chapter are available in the repository:

File	Description
<code>lgs_mgmttopchivalues.xlsx</code>	Top chi-square values
<code>lgs_mgmttopoddsvalues.xlsx</code>	Top odds ratios
<code>lgs_stepselect.xlsx</code>	Stepwise logistic results

III. Inputs

The chapter uses outputs from Chapter I:

Table	Description	Obs	Location
RAW TRAINING	Raw training data (no binning)	70,000	d:/lgs/lgs_rawTrain.sas7bdat
RAW HOLDOUT	Raw holdout data (no binning)	30,000	d:/lgs/lgs_rawHold.sas7bdat
LOGISTIC INPUT BINNED	Binned training data	70,000	d:/lgs/lgs_MgmAllChrNum.sas.
NORMALIZED CHAR BINNED	Character bins (long format)	980,000	d:/lgs/lgs_mgmNrmChr.sas7bd.
NORMALIZED NUM BINNED	Numeric bins (long format)	910,000	d:/lgs/lgs_mgmNrmNumSrt.sas.
CHR EXCEL REPORT	Character binning report	40	d:/lgs/lgs_mgmChrCutRpt
NUM EXCEL REPORT	Numeric binning report	46	d:/lgs/lgs_mgmNumCut

IV. Outputs

The primary output is the top 100 models for each model size (4-7 predictors):

Model Size	Number of Models
4 predictors 5 predictors 6	100
predictors 7 predictors Total	100
	100
	100
	400

Output location: d:/lgs/LGS_POSMGMMODNRM

V. Key Notes

1. This analysis works best with hundreds of independent variables. The methodology scales well with high-dimensional data.
2. It also works better if increasing values of the independent variables align with increasing response. Monotonic relationships improve model interpretability and performance.
3. Positive coefficients requirement. By default, all model predictors are required to have positive coefficients (intercept excluded). This provides:
 - More insight into the direction of effects
 - Easier interpretation of odds ratios
 - Models that "make sense" clinically or logically

The macro also supports identifying models with mixed sign coefficients to check for potentially better fits. Negative predictors often work against other predictors and may indicate multicollinearity or reverse coding.

4. All possible models with 4-7 independent variables are examined. The number of models tested is substantial:

```
n_choose_4 = comb(20, 4) = 4,845
n_choose_5 = comb(20, 5) = 38,760
n_choose_6 = comb(20, 6) = 38,760
n_choose_7 = comb(20, 7) = 77,520
Total = 159,885 models
```

Note: SLC does not support all-possible proc logistic models, so proc reg is used instead.

5. Stepwise logistic regression is performed to validate the best models identified by the all-possible search.

Additional Analysis:

- VIF (Variance Inflation Factor) - Variables with high VIF are removed and the model is rechecked
- Top Chi-Square Values - Identifies the most influential individual predictors
- Top Odds Ratios - Identifies predictors with the strongest effect sizes

CHAPTER III: VERIFYING TRAINING LOGISTIC USING HOLDOUT

Repository: utl-altair-slc-chapter-III-verifying-training-logistic-using-holdout

GitHub: <https://github.com/rogerjdeangelis/utl-altair-slc-chapter-III-verifying-training-logistic-using-holdout>

I. Best Model from Chapter II

The best model identified in Chapter II uses the following predictors:

- Response: `_MARRIED`
- Predictors:

- _ACSGENDER
- _DIVISIONCDE
- _INCOME
- _LOANHOME

II. Training Logistic

A logistic regression model is fitted on the training data (70,000 observations) using the five predictors identified above.

III. Create SAS Code to Bin Variables

The binning scheme from Chapter I is converted into SAS IF-THEN-ELSE code. This code is generated automatically using the binning cut points determined by the optimalbin macro on the training data.

IV. Map Character Variables

SAS IF-THEN-ELSE code is created to map character variables to their binned categories:

```
/* Example: Mapping character variable to bins */
IF _ACSGENDER IN ('M','Male') THEN _ACSGENDER_BIN = 1;
ELSE IF _ACSGENDER IN ('F','Female') THEN _ACSGENDER_BIN = 2;
/* ... etc. */
```

V. Map Numeric Variables

Similarly, SAS IF-THEN-ELSE code is created to map numeric variables to their binned categories:

```
/* Example: Mapping numeric variable to bins */
IF _INCOME < 25000 THEN _INCOME_BIN = 1;
ELSE IF 25000 <= _INCOME < 50000 THEN _INCOME_BIN = 2;
ELSE IF 50000 <= _INCOME < 75000 THEN _INCOME_BIN = 3;
ELSE _INCOME_BIN = 4;
```

VI. Apply IF-THEN-ELSE to Holdout

The generated IF-THEN-ELSE code is applied to the holdout sample (30,000 observations) to bin the data using the same cut points as the training data.

VII. Run Logistic on Holdout Sample

The logistic model (with the same coefficients estimated from the training data) is applied to the holdout sample to generate predicted probabilities.

VIII. Rank Probabilities

Predicted probabilities are ranked into deciles (10 groups) for both:

- Training sample predicted probabilities
- Holdout sample predicted probabilities

IX. Verify Training Logistic Using Holdout

The model is verified by comparing the lift (ratio of response rate in each decile to the overall response rate) between training and holdout samples.

X. Compare Lift Plots - Training and Holdout

A lift plot comparison is generated:

<https://github.com/rogerjdeangelis/utl-altair-slc-chapter-III-verifying-training-logistic-using-holdout/blob/main/liftplots.png>

XI. ASCII Plot Verification

The differences between training and holdout probabilities are small (most < 0.01), indicating that the model generalizes well to new data.

XII. Output Datasets

The chapter produces the following output datasets:

Dataset	Description
Training probabilities	Predicted probabilities for training sample, by decile
Holdout probabilities	Predicted probabilities for holdout sample, by decile
Lift comparison	Side-by-side lift values for training and holdout

Key Finding: The holdout validation confirms that the logistic model developed on the training data performs consistently on unseen data, with minimal degradation in predictive performance.

CHAPTER IV: KEY OUTPUTS FROM THE FINAL LOGISTIC MODEL

Repository: `utl-altair-slc-chapter-IV-key-outputs-from-the-final-logistic-model`

GitHub: <https://github.com/rogerjdeangelis/utl-altair-slc-chapter-IV-key-outputs-from-the-final-logistic-model>

I. Key Outputs Overview

This chapter presents the final outputs from the logistic regression model, including:

- Gains and lift tables
- ROC curves
- Relative contribution charts
- Comprehensive logistic analysis tables (14 Excel tabs)
- Publishable PDF reports

II. Gains Lift Table

The gains lift table provides a comprehensive view of model performance:

File: lgs_ganrptfin.xlsx

https://github.com/rogerjdeangelis/utl-altair-slc-chapter-IV-key-outputs-from-the-final-logistic-model/blob/main/lgs_ganrptfin.xlsx

The table includes:

- Decile (1-10, from highest to lowest predicted probability)
- Number of observations per decile
- Number of responders per decile
- Response rate per decile
- Cumulative response rate
- Lift (response rate in decile / overall response rate)
- Cumulative lift

III. ROC Curves

ROC (Receiver Operating Characteristic) curves are provided for:

- The overall model (overlay of all predictors)
- Individual predictors:
 - MARRIED ROC curve
 - ACSGENDER ROC curve
 - DIVISIONCDE ROC curve
 - INCOME ROC curve
 - LOANHOME ROC curve

File: ROCOverlay.png

<https://github.com/rogerjdeangelis/utl-altair-slc-chapter-IV-key-outputs-from-the-final-logistic-model/blob/main/ROCOverlay.png>

IV. Perfect Relative Contributions to Max Probability

A pie chart showing the relative contribution of each predictor to the maximum probability score:

File: Relative_Contributions.png (also SGPie.png)

<https://github.com/rogerjdeangelis/utl-altair-slc-chapter-IV-key-outputs-from-the-final-logistic-model/blob/main/SGPie.png>

This visualization helps identify which variables are driving the highest predicted probabilities.

V. Full Logistic Analysis Tables

Comprehensive logistic analysis is provided in multiple formats:

Format	File	Description
--------	------	-------------

Excel	logistic_tables.xlsx	14 Excel tabs with detailed logistic analysis
HTML	logistic_tables.html	Web-compatible version
PDF	logistic_tables.pdf	Publishable version
RTF	logistic_tables.rtf	Microsoft Word compatible

https://github.com/rogerjdeangelis/utl-altair-slc-chapter-IV-key-outputs-from-the-final-logistic-model/blob/main/logistic_tables.xlsx

VI. Inputs

Only one input is required for this repository - the outputs from Chapters I and II are used as inputs.

VII. Outputs

All outputs are in the repository:

```
d:/lgs
+---htm
|  logistic_tables.html
+---pdf
|  logistic_tables.pdf
+---sas
|  slc_gain.sas          (Gains and Lift table code, Word-compatible)
|  slc_pdflan100.sas     (Proc template for PDF output)
|  slc_rtfan100.sas     (Proc template for RTF output)
+---rtf
|  logistic_tables.rtf
+---png
|  Relative_Contributions.png
|  ROCOverlay.png
|  ROCurve1.png
|  SGPie.png
\---xls
   lgs_5best.xlsx        (Best model from 136,789 models examined)
   lgs_ganrptfin.xlsx    (Gains Lift Table)
   lgs_MgmChrCut.xlsx     (Best Character Bins)
   lgs_MgmNumCut.xlsx     (Best Numeric Bins)
   lgs_mgmtopchivalues.xlsx (Chi-Square analysis -- Mantel-Haenszel linear test)
   lgs_mgmtopoddsvalues.xlsx (Odds Ratio analysis)
   lgs_stepselect.xlsx    (Stepwise logistic -- validates best models)
```

Summary of Key Files:

File	Description
------	-------------

lgs_5best.xlsx	lgs_ganrptfin.xlsx	Best model from 136,789 models examined
lgs_MgmChrCut.xlsx	lgs_MgmNumCut.xlsx	Gains and Lift Table
lgs_mgmtopchivalues.xlsx		Best Character Bins
lgs_mgmtopoddsvalues.xlsx	lgs_stepselect.xlsx	Best Numeric Bins
		Chi-Square analysis (Mantel-Haenszel linear test)
		Odds Ratio analysis
		Stepwise logistic results

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End of ebook
